Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An object weighs **29.4 N** where g=**9.8 N/kg**. **Find its mass** and its weight in a field of **5 N/kg**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A **2 kg** test object feels **6 N** toward a source. The local field is g=**6/2**=**3 N/kg**. A **4 kg** test object at that point feels **12 N**.

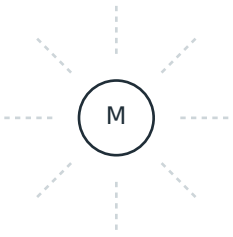
Your turn: 1. **Draw the field arrow** and the two possible force arrows.
2. Which quantity belongs to the location rather than the test mass?

Test mass (kg)	Force (N)	g (N/kg)
2	6	____
4	12	____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw field arrows** at three locations around the source.
2. Which way would a small test mass accelerate?



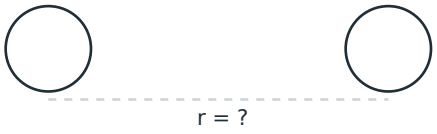
Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Apply the physics

A **0.20 kg** test mass experiences a **0.60 N** gravitational force. **Find the field strength**. The source has GM = **12 m³/s²**; **find the test mass's center-to-center** distance from the source.



Source: GM = 12 Test mass: 0.20 kg

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force. Use F=mg and a=F/m to respond.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

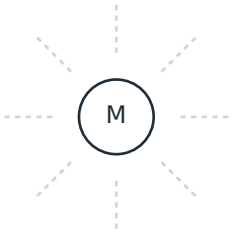
A **4 kg** block slides down a **30°** ramp with **11.6 N** friction up the ramp. Use g=**9.8 m/s²**. **Find along-ramp acceleration** and normal force.
Draw the forces on the chosen object and **label their sources**.

Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

Outside a spherical source, field is **8 N/kg** at r. **Find field** at **2** r and force on **0.50 kg** there.



Add arrowheads to show field direction.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

An exterior field is **8 N/kg** at distance r from a spherical source. Where is it **2 N/kg**? **What force** acts there on a **0.25 kg** mass?

Location	Test mass (kg)	Field (N/kg)	Force (N)
1	___	___	___
2	___	___	___

Marzano check - circle: ① ② ③